

University of Dhaka



Assignment - 1

Course Title: Advanced Econometrics-II
Course Code: 408

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Question:

Using Google Finance, download the data on the Bangladeshi Taka to the United States Dollar exchange rate (hint: first use Google Sheets to retrieve the data) from 1 January 2011 to 1 January 2026.

Task 1: Import the data to STATA

Task 2: Draw a line graph with the exchange rate on the Y-axis and time on the X-axis.

Task 3: Create a new variable **Diff_Ex**. The variable should include the first difference values of the exchange rate.

Task 4: Suggest what univariate ARMA model **the change in the exchange** rate follows.

Write the report on all these tasks. Upload the report and the STATA code as a single PDF.

Solutions:

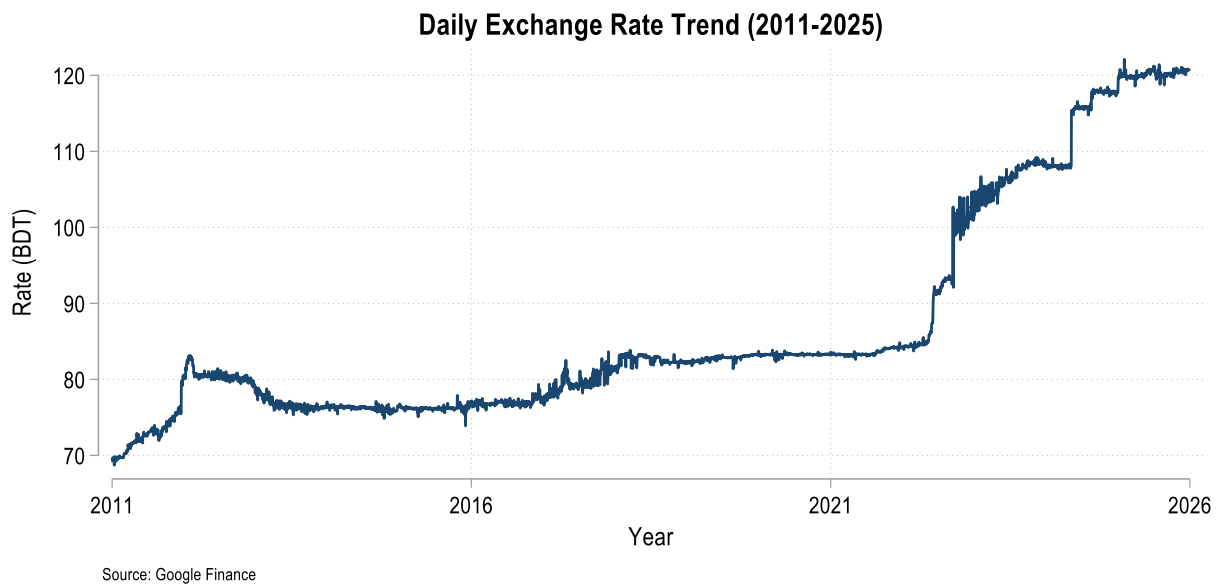
Task-1:

First, I have used Google finance to retrieve the USD to BDT exchange rate from 1 January 2011 to 1 January 2026, total 15 years of data. For this task I have used this formula,

```
=GOOGLEFINANCE("CURRENCY:USDBDT","price",DATE(2011,1,1),DATE(2026,1,1),"Daily")
```

There are 4480 daily observations starting from 3rd January 2011 to 31st December 2025. It excludes all the business holidays on Saturdays and Sundays. For calculation simplicity in Stata, I have converted the date format to an internal business calendar using Stata's *bcal* command. This will ensure that all the calculations acknowledge this time gap in date time column of the data.

Task-2:



At the start of the year 2011 the exchange rate was close to 70 BDT. Later it becomes 86 BDT and stayed unchanged with no trend for approximately 10 years. After the Covid-19 outbreak the exchange rate showed a huge upward trend from 86 BDT to 120 BDT and at the end of year 2025 it is consistent with the trend.

Task-3:

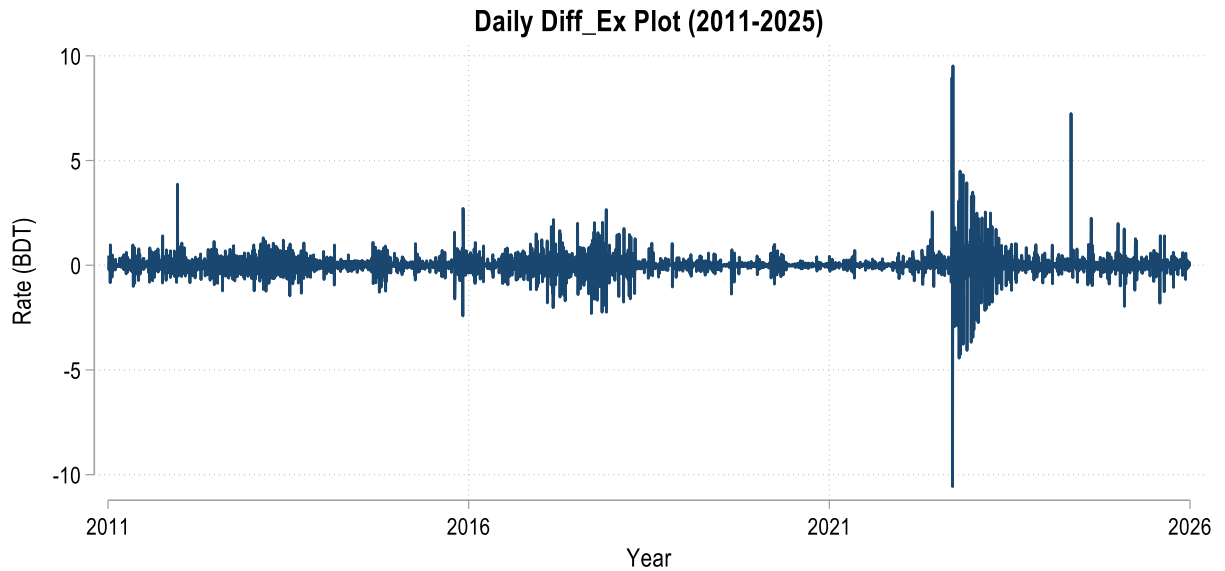
I used Stata's *generate Diff_Ex = D.Ex_Rate* command to create the first Difference of the Exchange Rate. Before that I converted the Date column into a business calendar to avoid the calculation miscreancy.

Task-4:

Ans: ARMA (3,2) Model.

Following the Box-Jenkins Methodology, I will follow (a) Identification (b) Estimation (c) Diagnostic, these 3 procedures to find the best Univariate ARMA model that the difference of the exchange rate follows.

(a) Identification

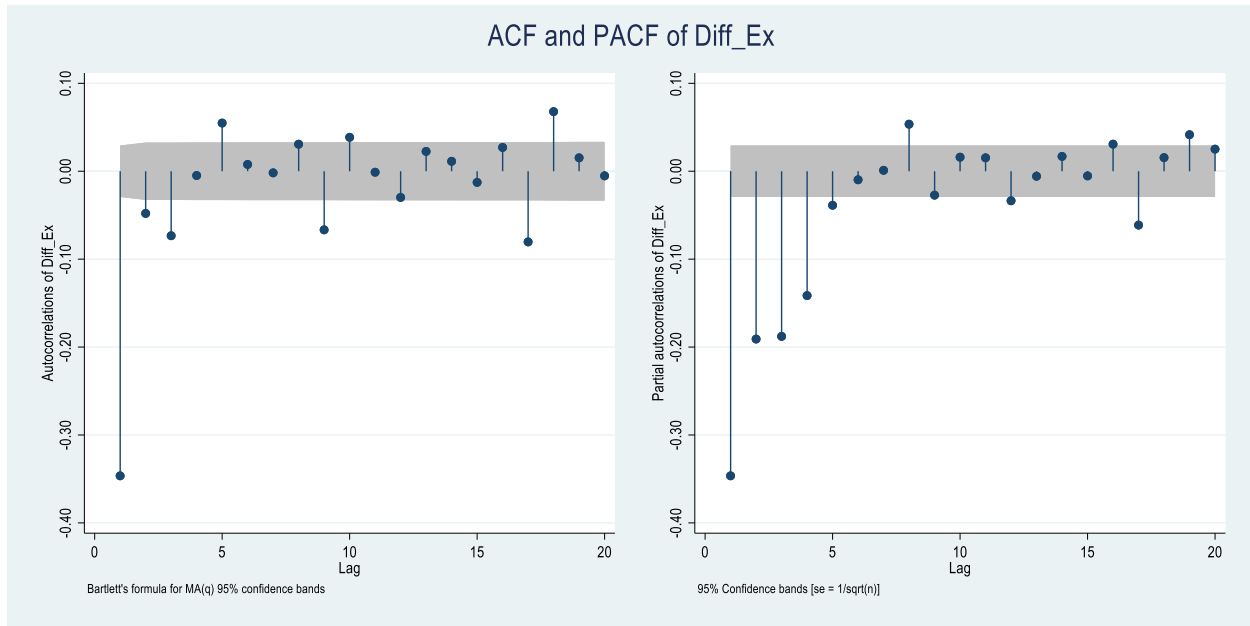


H0: Random walk without drift, $d = 0$

	Test	Dickey Fuller Critical Values		
	Statistics	1%	5%	10%
Z(t)	-84.872	-3.430	-2.860	-2.570

MacKinnon approximate p-value for $Z(t) = 0.0000$.

The Time Series plot of the first difference of exchange rate clearly shows no significant change in mean and no trend. And the Dicky-Fuller Test also rejects the Null Hypothesis of an Unit Root Process. So, the Diff_Ex series is stationary.



Both the Autocorrelation and Partial Auto-correlation Function of Diff_Ex variables show significant spikes till lag 3 then exponential decay. So, I assume the presence of ARMA (2,3), ARMA (3,2), ARMA (3,3), ARMA (3,4), ARMA (4,3). Now I will estimate all the assumed models and compare their information criteria to get the parsimonious model.

Information Criteria:

<i>Model</i>	<i>N</i>	<i>LL (model)</i>	<i>df</i>	<i>AIC</i>	<i>BIC</i>
ARMA (2,3)	4479	-3029.567	7	6073.134	6117.984
ARMA (3,2)	4479	-3028.1	7	6070.201	6115.051
ARMA (3,3)	4479	-3027.874	8	6071.749	6123.006
ARMA (3,4)	4479	-3027.815	9	6073.63	6131.295
ARMA (4,3)	4479	-3027.432	9	6072.865	6130.529

Among those guessed model , ARMA (3,2) yields the lowest AIC and BIC so this is my chosen Univariate model for the first difference of exchange rate.

(b) Estimation:

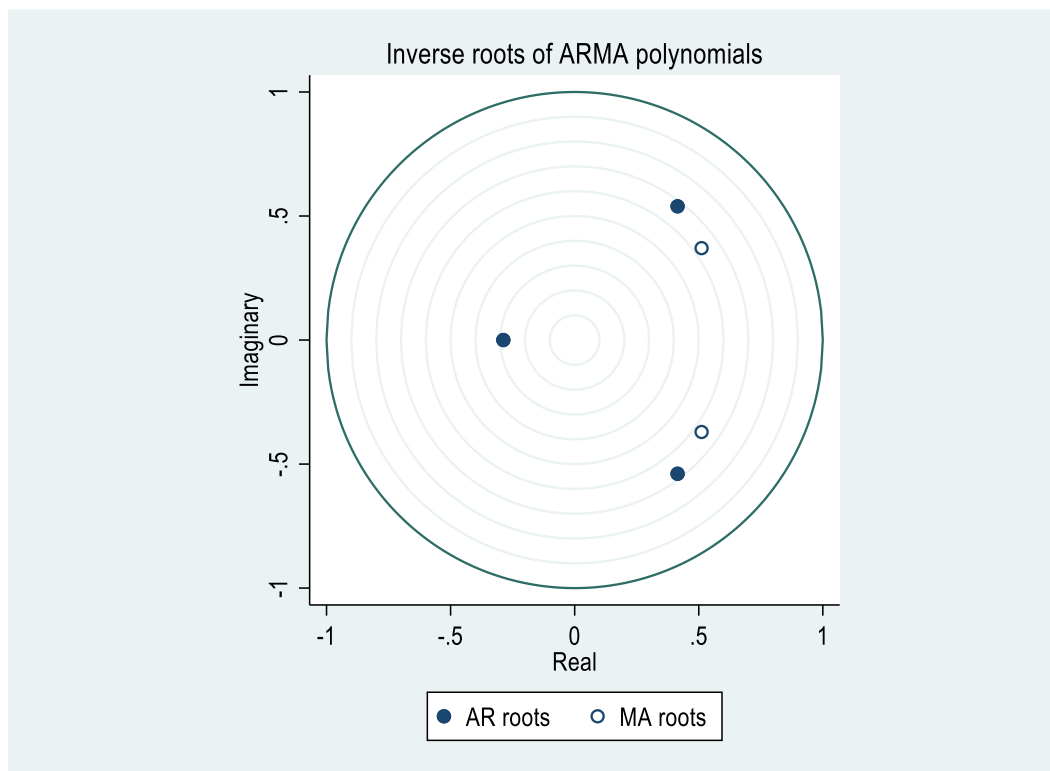
ARIMA (3,2) regression

Diff_Ex	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
Constant	.011	.004	2.89	.004	.004	.019	***
ar							
L	.542	.065	8.29	0	.414	.67	***
L2	-.224	.018	-12.33	0	-.26	-.188	***
L3	-.133	.011	-11.92	0	-.155	-.111	***
ma							
L	-1.023	.066	-15.49	0	-1.152	-.893	***
L2	.399	.046	8.58	0	.308	.49	***
Constant	.476	.001	465.0	0	.474	.478	***

2

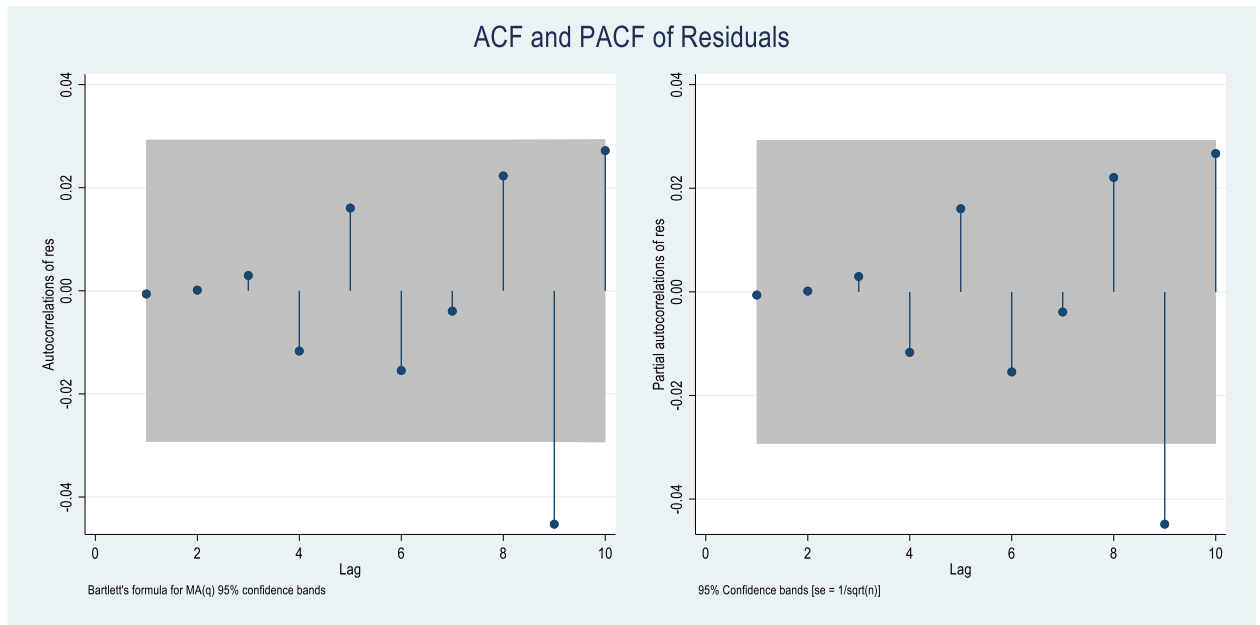
Mean dependent var	0.012	SD dependent var	0.532
Number of obs	4479	Chi-square	98684.671
Prob > chi2	.	Akaike crit. (AIC)	6070.201

*** $p < .01$, ** $p < .05$, * $p < .1$



After estimating the ARMA (3,2) model we can see all the lag coefficients of AR and MA process are statistically significant at 1% confidence intervals. And the roots diagram shows us that all the roots of AR and MA process are within the unit circle. So, we can say ARMA (3,2) model is acceptable for estimation.

(c) Diagnostic Test:



Portmanteau test for white noise

$$\text{Portmanteau (Q) statistics} = 17.7000$$

$$\text{Prob} > \text{chi2}(10) = 0.0602$$

The Ljung-Box test or Q test at lag 10 do not reject the Null Hypothesis of the residuals being white noise. The Autocorrelation and Partial Autocorrelation function of the residuals also justify this white noise process for lag 10. This diagnostic test satisfies the residuals being white noise and according to the steps of Box Jenkins Methodology ARMA (3,2) model is accepted as the univariate process of the difference of the exchange rate.

Ans: ARMA (3,2) model.

```

1  cd "F:\ECON 408\Practice do file"
2
3
4  clear all
5  set seed 1000
6
7  * Task 1:
8  *-----
9
10
11 import excel "USD to BDT.xlsx", firstrow clear
12
13 rename Close Ex_Rate
14 label variable Ex_Rate "Exchange Rate"
15
16 replace Date = dofc(Date)
17 format Date %td
18 tsset Date
19
20
21 * Task 2:
22 *-----
23
24
25 set scheme cleanplots
26
27 tsline Ex_Rate, ///
28         lcolor(navy) lwidth(medthick) ///
29         title("{bf:Daily Exchange Rate Trend (2011-2025)}",
30 size(large)) ///
31         xtitle("Year", size(medlarge)) ///
32         ytitle("Rate (BDT)", size(medlarge)) ///
33         ylabel(, labsize(medium)) ///
34         tlabel(01jan2011 01jan2016 01jan2021 01jan2026,
35 format(%tdcY) labsize(medium)) ///
36         graphregion(color(white)) xsize(8) ysize(4) ///
37         name(tsplot, replace) note("Source: Google Finance")
38
39 graph export Hw_1_tsplot.emf, name(tsplot) replace

```



```

36
37
38 * Task 3:
39 *-----
-----
40
41 bcal create bus_cal, from(Date) gen(time) replace
42 format time %tbbus_cal
43 tsset time
44 gen Diff_Ex = D.Ex_Rate
45
46
47 * Task 4:
48 *-----
-----
49
50 * Step 1: Stationarity Check
51
52 tsset Date
53
54 set scheme cleanplots
55
56 tsline Diff_Ex, ///
57         lcolor(navy) lwidth(medthick) ///
58         title("{bf:Daily Diff_Ex Plot (2011-2025)}", size(
59 large)) ///
60         xtitle("Year", size(medlarge)) ///
61         ytitle("Rate (BDT)", size(medlarge)) ///
62         ylabel(, labsize(medium)) ///
63         tlabel(01jan2011 01jan2016 01jan2021 01jan2026,
64 format(%tdcY) labsize(medium)) ///
65         graphregion(color(white)) xsize(8) ysize(4) ///
66         name(Diff_Ex_plot, replace)
67
68 graph export Diff_Ex_plot.emf, name(Diff_Ex_plot) replace
69
70 dfuller Diff_Ex
71 tsset time
72

```

```

71 * Step 2: Empirical ACF and PACF check
72 set scheme s2color
73
74 ac Diff_Ex, name(acf_plot,replace) lags(20)
75 pac Diff_Ex, name(pacf_plot,replace) lags(20)
76 graph combine acf_plot pacf_plot, title("ACF and PACF of
Diff_Ex") xsize(8) ysize(4) name(acf_pacf_plot, replace)
77
78 graph export hw_1_acf_pacf_plot.emf, name(acf_pacf_plot)
replace
79
80 *ARMA(2,3) , ARMA(3,2) , ARMA(3,3) , ARMA(3,4), ARMA(4,3)
81
82
83 qui arima Diff_Ex, ar(1/2) ma(1/3)
84 estat ic
85
86 qui arima Diff_Ex, ar(1/3) ma(1/2)
87 estat ic
88
89 qui arima Diff_Ex, ar(1/3) ma(1/3)
90 estat ic
91
92 qui arima Diff_Ex, ar(1/3) ma(1/4)
93 estat ic
94
95 qui arima Diff_Ex, ar(1/4) ma(1/3)
96 estat ic
97
98
99 * ARMA(3,2) Model has the lowest AIC and BIC
100
101 * Step 3: Estimation
102
103 arima Diff_Ex, ar(1/3) ma(1/2)
104 estat ic
105 estat aroots , name(arootshw1, replace)
106 graph export arootshw1.emf, name(arootshw1) replace
107 predict res , residuals

```

```
108
109 ac res, name(acf_res,replace) lags(10)
110 pac res, name(pacf_res,replace) lags(10)
111 graph combine acf_res pacf_res, title("ACF and PACF of
Residuals") xsize(8) ysize(4) name(hw1_res_ac_pac_plot,
replace)
112 graph export hw1_res_ac_pac_plot.emf, name(
hw1_res_ac_pac_plot) replace
113
114
115 wntestq res, lags(10)
116
117
118
```